3.2: Four emergent properties of water contribute to Earth's suitability for life

We will examine four emergent properties of water that contribute to Earth's suitability as an environment for life: cohesive behavior, ability to moderate temperature, expansion upon freezing, and versatility as a solvent.

Cohesion of Water Molecules

Water molecules stay close to each other as a result of hydrogen bonding. Although the arrangement of molecules in a sample of liquid water is constantly changing, at any given moment many of the molecules are linked by multiple hydrogen bonds. These linkages make water more structured than most other liquids. Collectively, the hydrogen bonds hold the substance together, a phenomenon called **cohesion**.

One result of cohesion due to hydrogen bonding is high **surface tension**, a measure of how difficult it is to stretch or break the surface of a liquid. At the air-water interface is an ordered arrangement of water molecules, hydrogen-bonded to one another and to the water below, but not to the air above. This asymmetry gives water an unusually high surface tension, making it behave as though it were coated with an invisible film. The spider in **Figure 3.3** takes advantage of the surface tension of water to walk across a pond without breaking the surface, and some plants can float on water as well. You can observe the surface tension of water by slightly overfilling a drinking glass; the water will stand above the rim.

Figure 3.3

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Walking on water.
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A raft spider rests on the surface of a body of water.

The high surface tension of water, resulting from the collective strength of its hydrogen bonds, allows this raft spider to walk on the surface of a pond.

Cohesion also contributes to the transport of water and dissolved nutrients against gravity in plants **(Figure 3.4)**. Water from the roots reaches the leaves through a network of water-conducting cells. As water evaporates from a leaf, hydrogen bonds cause water molecules leaving the veins to tug on molecules farther down, and the upward pull is transmitted through the water-conducting cells all the way to the roots.

Adhesion, the clinging of one substance to another, also plays a role. Adhesion of water by hydrogen bonds to the molecules of cell walls helps counter the downward pull of gravity (see Figure 3.4).

Figure 3.4

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Water transport in plants.
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Water transport in plants.

A tree is diagrammed. The direction of water movement goes from the H 2 O in the ground and up through the tree trunk, then out to the branches. Evaporation from leaves pulls water upward from the roots through water-conducting cells; there are two types of water-conducting cells. Adhesion of the water to cell walls by hydrogen bonds helps resist the downward pull of gravity. Cohesion due to hydrogen bonds between water molecules helps hold together the column of water within the cells.

Evaporation from leaves pulls water upward from the roots through water-conducting cells. Because of the properties of cohesion and adhesion, the tallest trees can transport water more than 100 m upward—approximately one-quarter the height of the Empire State Building in New York City.

BioFlix® Animation

Adhesion and Cohesion in Plants

image

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image

Play

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Animation

Cohesion of Water

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Moderation of Temperature by Water

Water moderates air temperature by absorbing heat from air that is warmer and releasing stored heat to air that is cooler. Water is effective as a heat bank because it can absorb or release a relatively large amount of heat with only a slight change in its own temperature. To understand this capability of water, let's first look at temperature and heat.

Temperature and Heat

Anything that moves has **kinetic energy**, the energy of motion. Atoms and molecules have kinetic energy because they are always moving, although not necessarily in any particular direction. The faster a molecule moves, the greater its kinetic energy. The kinetic energy associated with the random movement of atoms or molecules is called **thermal energy**. Thermal energy is related to temperature, but they are not the same thing. **Temperature** represents the *average* kinetic energy of the molecules in a body of matter, regardless of volume, whereas the thermal energy of a body of matter reflects the *total* kinetic energy, and thus depends on the matter's volume. When water is heated in a coffeemaker, the average speed of the molecules increases, and the thermometer records this as a rise in temperature of the liquid. The total amount of thermal energy also increases in this case. Note, however, that although the pot of coffee has a much higher temperature than, say, the water in a swimming pool, the swimming pool contains more thermal energy because of its much greater volume.

Whenever two objects of different temperature are brought together, thermal energy passes from the warmer to the cooler object until the two are the same temperature. Molecules in the cooler object speed up at the expense of the thermal energy of the warmer object. An ice cube cools a drink not by adding coldness to the liquid but by absorbing thermal energy from the liquid as the ice itself melts. Thermal energy in transfer from one body of matter to another is defined as **heat**.

One convenient unit of heat used in this book is the **calorie (cal)**. A calorie is the amount of heat it takes to raise the temperature of 1g of water by 1°C. Conversely, a calorie is also the amount of heat that 1g of water releases when it cools by 1°C. A **kilocalorie (kcal)**, 1,000 cal, is the quantity of heat required to raise the temperature of 1 kilogram (kg) of water by 1°C. (The "Calories" on food packages are actually kilocalories.) Another energy unit used in this book is the **joule (J)**. One joule equals 0.239 cal; one calorie equals 4.184 J.

Water's High Specific Heat

The ability of water to stabilize temperature stems from its relatively high specific heat. The **specific heat** of a substance is defined as the amount of heat that must be absorbed or lost for 1 g of that substance to change its temperature by 1°C. We already know water's specific heat because we have defined a calorie as the amount of heat that causes 1 g of water to change its temperature by 1°C. Therefore, the specific heat of water is 1 calorie per gram and per degree Celsius, abbreviated as 1 cal/(g·°C). Compared with most other substances, water has an unusually high specific heat. For example, ethyl alcohol, the type of alcohol in alcoholic beverages, has a specific heat of 0.6 cal/(g·°C); that is, only 0.6 cal is required to raise the temperature of 1 g of ethyl alcohol by 1°C.

Because of the high specific heat of water relative to other materials, water will change its temperature less than other liquids when it absorbs or loses a given amount of heat. The reason you can burn your fingers by touching the side of an iron pot on the stove when the water in the pot is still lukewarm is that the specific heat of water is ten times greater than that of iron. In other words, the same amount of heat will raise the temperature of 1 g of the iron much faster than it will raise the temperature of 1 g of the water. Specific heat can be thought of as a measure of how well a substance resists changing its temperature when it absorbs or releases heat. Water resists changing its temperature; when it does change its temperature, it absorbs or loses a relatively large quantity of heat for each degree of change.

We can trace water's high specific heat, like many of its other properties, to hydrogen bonding. Heat must be absorbed in order to break hydrogen bonds; by the same token, heat is released when hydrogen bonds form. A calorie of heat causes a relatively small change in the temperature of water because much of the heat is used to disrupt hydrogen bonds before the water molecules can begin moving faster. And when the temperature of water drops slightly, many additional hydrogen bonds form, releasing a considerable amount of energy in the form of heat.

What is the relevance of water's high specific heat to life on Earth? A large body of water can absorb and store a huge amount of heat from the sun in the daytime and during summer while warming up only a few degrees. At night and during winter, the gradually cooling water can warm the air. This capability of water serves to moderate air temperatures in coastal areas **(Figure 3.5)**. The high specific heat of water also tends to stabilize ocean temperatures, creating a favorable environment for marine life. Thus, because of its high specific heat, the water that covers most of Earth keeps temperature fluctuations on land and in water within limits that permit life. Also, because organisms are made primarily of water, they are better able to resist changes in their own temperature than if they were made of a liquid with a lower specific heat.

Figure 3.5

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    Temperatures for the Pacific Ocean and Southern California
on an August day.
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Pacific Ocean is 68 degrees Fahrenheit. Temperatures are cooler at the coast (low 70s) and increase as you travel inland (Palm Springs is 106 degrees).

Pacific Ocean is 68 degrees Fahrenheit. Temperatures in the coastal cities range from 73 degrees (Santa Barbara), 75 degrees (Los Angeles), to 72 degrees (San Diego). Inland temperatures increase the further inland you move. Burbank (90 degrees) and Santa Ana (84 degrees) are both approximately 20 miles from the coast. San Bernardino (100 degrees), Riverside (96 degrees) and Palm Springs (106 degrees) are 40 to 50 miles from the coast, respectively.

INTERPRET THE DATA *Explain the pattern of temperatures shown in this diagram.*

Evaporative Cooling

Molecules of any liquid stay close together because they are attracted to one another. Molecules moving fast enough to overcome these attractions can depart the liquid and enter the air as a gas (vapor). This transformation from a liquid to a gas is called vaporization, or evaporation. Recall that the speed of molecular movement varies and that temperature is the *average* kinetic energy of molecules. Even at low temperatures, the speediest molecules can escape into the air. Some evaporation occurs at any temperature; a glass of water at room temperature, for example, will eventually evaporate completely. If a liquid is heated, the average kinetic energy of molecules increases and the liquid evaporates more rapidly.

Heat of vaporization is the quantity of heat a liquid must absorb for 1 g of it to be converted from the liquid to the gaseous state. For the same reason that water has a high specific heat, it also has a high heat of vaporization relative to most other liquids. To evaporate 1 g of water at 25°C, about 580 cal of heat is needed —nearly double the amount needed to vaporize a gram of alcohol or ammonia. Water's high heat of vaporization is another emergent property resulting from the strength of its hydrogen bonds, which must be broken before the molecules can exit from the liquid in the form of water vapor.

The high amount of energy required to vaporize water has a wide range of effects. On a global scale, for example, it helps moderate Earth's climate. A considerable amount of solar heat absorbed by tropical seas is consumed during the evaporation of surface water. Then, as moist tropical air circulates poleward, it releases heat as it condenses and forms rain. On an organismal level, water's high heat of vaporization accounts for the severity of steam burns. These burns are caused by the heat energy released (during formation of hydrogen bonds) when steam condenses into liquid on the skin.

As a liquid evaporates, the surface of the liquid that remains behind cools down (its temperature decreases). This **evaporative cooling** occurs because the “hottest” molecules, those with the greatest kinetic energy, are the most likely to leave as gas. It is as if the 100 fastest runners at a college transferred to another school; the average speed of the remaining students would decline.

Evaporative cooling of water contributes to the stability of temperature in lakes and ponds and also provides a mechanism that prevents terrestrial organisms from overheating. For example, evaporation of water from the leaves of a plant helps keep the tissues in the leaves from becoming too warm in the sunlight. Evaporation of sweat from human skin dissipates body heat and helps prevent overheating on a hot day or when excess heat is generated by strenuous activity. High humidity on a hot day increases discomfort because the high concentration of water vapor in the air inhibits the evaporation of sweat from the body. Animals without sweat glands, such as elephants, may spray water on themselves to cool down **(Figure 3.6)**.

Figure 3.6

Evaporative cooling.

An elephant sprays itself with water. Water molecules with the greatest kinetic energy leave as gas, making the remaining water cooler.

In hot weather, an elephant sprays water from its trunk onto its head. Evaporation of this water cools the elephant down.

Floating of Ice on Liquid Water

Water is one of the few substances that are less dense as a solid than as a liquid. As a result, ice floats on liquid water. While other materials contract and become denser when they solidify, water expands. The cause of this exotic behavior is, once again, hydrogen bonding. At temperatures above 4°C, water behaves like other liquids, expanding as it warms and contracting as it cools. As the temperature falls from 4°C to 0°C, water begins to freeze because more and more of its molecules are moving too slowly to break hydrogen bonds. At 0°C, the molecules become locked into a crystalline lattice, each water molecule hydrogen-bonded to four partners (see Figure 3.1). The hydrogen bonds keep the molecules at “arm’s length,” far enough apart to make ice about 10% less dense 10% fewer molecules in the same volume) than liquid water at 4°C. When ice absorbs enough heat for its temperature to rise above 0°C, hydrogen bonds between molecules are disrupted. As the crystal collapses, the ice melts and molecules have fewer hydrogen bonds, allowing them to slip closer together. Water reaches its greatest density at 4°C and then begins to expand as the molecules move faster. Even in liquid water, many of the molecules are connected by hydrogen bonds, though only transiently: The hydrogen bonds are constantly breaking and re-forming.

The ability of ice to float due to its lower density is an important factor in the suitability of the environment for life. If ice sank, then eventually ponds, lakes, and even oceans could freeze solid, making life as we know it impossible on Earth. During summer, only the upper few inches of the ocean would thaw. Instead, when a deep body of water cools, the ice floats, insulating the liquid water below. This prevents it from freezing and allows life to exist under the frozen surface, as shown in Figure 3.1. Besides insulating the water below, ice also provides a solid habitat for some animals, such as polar bears and seals.

Many scientists are worried that these bodies of ice are at risk of disappearing. Global warming, which is caused by carbon dioxide and other "greenhouse" gases in the atmosphere (see Figure 56.30), is having a profound effect on icy environments around the globe. In the Arctic, the average air temperature has risen 2.2°C just since 1961. This temperature increase has affected the seasonal balance between Arctic sea ice and liquid water, causing ice to form later in the year, to melt earlier, and to cover a smaller area. The rate at which glaciers and Arctic sea ice are disappearing is posing an extreme challenge to animals that depend on ice for their survival **(Figure 3.7)**.

Figure 3.7

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Effects of climate change on the Arctic.
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Labeled diagram shows effects of climate change on the Arctic.

Warmer temperatures in the Arctic cause more sea ice to melt in the summer. The loss of ice disrupts the ecosystem, affecting many species. (Map data from the National Snow and Ice Data Center.)

Top-left corner of the diagram shows a globe. The remainder of the diagram shows an expanded view of a map of the Arctic region. The map shows the Arctic Ocean, Russia, Alaska, Canada, the Arctic ice sheet and Greenland, the Bering Strait, and the North Pole. A red line shows the median extent of the ice sheet in September 1981 to 2010. It goes beyond Greenland and almost close to Russia, Alaska and Canada. The extent of the ice sheet in September 2019 is much smaller.

Species harmed by loss of Arctic ice:

- Less ice reduces feeding opportunities for polar bears, who hunt from the ice.
- Black guillemots in Alaska cannot fly from their nests on land to their fishing grounds at the edge of the ice, which is now too far from land; young birds are starving.
- Loss of floating ice as habitat has caused a decline in Pacific walrus populations due to overcrowding and deadly stampedes on land.

Species benefiting from loss of Arctic ice:

- Warm water and more light result in more phytoplankton, which are eaten by other organisms. Harmful algal blooms are also a threat.
- Populations of bowhead whales and some fish species may be increasing because more plankton are available to eat.

Interview

Interview with Susan Solomon: Understanding climate change

A photo of Susan Solomon.

Water: The Solvent of Life

A sugar cube placed in a glass of water will dissolve. In time, the glass will contain a uniform mixture of sugar and water; the concentration of dissolved sugar will be the same everywhere in the mixture. A liquid that is a completely homogeneous mixture of two or more substances is called a **solution**. The dissolving agent of a solution is the **solvent**, and the substance that is dissolved is the **solute**. In this case, water is the solvent and sugar is the solute. An **aqueous solution** is one in which the solute is dissolved in water; water is the solvent.

Water is a very versatile solvent, a quality we can trace to the polarity of the water molecule. Suppose, for example, that a spoonful of table salt, the ionic compound sodium chloride (NaCl), is placed in water **(Figure 3.8)**. At the surface of each crystal (grain) of salt, the sodium and chloride ions are exposed to the solvent. These ions and regions of the water molecules are attracted to each other due to their opposite charges. The oxygens of the water molecules have regions of partial negative charge that are attracted to sodium cations. The hydrogen regions are partially positively charged and are attracted to chloride anions. As a result, water molecules surround the individual sodium and chloride ions, separating and shielding them from one another. The sphere of water molecules around each dissolved ion is called a **hydration shell**. Working inward from the surface of each salt crystal, water eventually dissolves all the ions. The result is a solution of two solutes, sodium cations and chloride anions, mixed homogeneously with water, the solvent. Other ionic compounds also dissolve in water. Seawater, for instance, contains a great variety of dissolved ions, as do living cells.

Figure 3.8

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Table salt dissolving in water.
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Negative oxygen regions of polar water molecules are attracted to sodium cations (N A plus). Positive hydrogen regions of water molecules are attracted to chloride anions (C l minus).

A sphere of water molecules, called a hydration shell, surrounds each solute ion.

WHAT IF? *What would happen if you heated this solution for a long time?*

*For suggested answer, see**Appendix A.*

A compound does not need to be ionic to dissolve in water; many compounds made up of nonionic polar molecules, such as the sugar in the sugar cube mentioned earlier, are also water-soluble. Such compounds dissolve when water molecules surround each of the solute molecules, forming hydrogen bonds with them. Even molecules as large as proteins can dissolve in water if they have ionic and polar regions on their surface **(Figure 3.9)**. Many different kinds of polar compounds are dissolved (along with ions) in the water of such biological fluids as blood, the sap of plants, and the liquid within all cells. Water is the solvent of life.

Figure 3.9

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A water-soluble protein.
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Oxygen is attracted to a slight positive charge on the lysozyme molecule. Hydrogen is attracted to a slight negative charge on the lysozyme molecule.

Human lysozyme is a protein found in tears and saliva that has antibacterial action (see Figure 5.16). This model shows the lysozyme molecule (purple) in an aqueous environment. Ionic and polar regions on the protein's surface attract the partially charged regions on water molecules.

Hydrophilic and Hydrophobic Substances

Any substance that has an affinity for water is said to be **hydrophilic** (from the Greek *hydro*, water, and *philos*, loving). In some cases, substances can be hydrophilic without actually dissolving. For example, some molecules in cells are so large that they do not dissolve. Another example of a hydrophilic substance that does not dissolve is cotton, a plant product. Cotton consists of giant molecules of cellulose, a compound with numerous regions of partial positive and partial negative charges that can form hydrogen bonds with water. Water adheres to the cellulose fibers. Thus, a cotton towel does a great job of drying the body, yet it does not dissolve in the washing machine. Cellulose is also present in the walls of water-conducting cells in a plant; you read earlier how the adhesion of water to these hydrophilic walls helps water move up the plant against gravity.

There are, of course, substances that do not have an affinity for water. Substances that are nonionic and nonpolar (or otherwise cannot form hydrogen bonds) actually seem to repel water; these substances are said to be **hydrophobic** (from the Greek *phobos*, fearing). An example from the kitchen is vegetable oil, which, as you know, does not mix stably with water-based substances such as vinegar. The hydrophobic behavior of the oil molecules results from a high number of relatively nonpolar covalent bonds, in this case bonds between carbon and hydrogen, which share electrons almost equally. Hydrophobic molecules related to oils are major ingredients of cell membranes. (Imagine what would happen to a cell if its membrane dissolved!)

Solute Concentration in Aqueous Solutions

Most of the chemical reactions in organisms involve solutes dissolved in water. To understand such reactions, we must know how many atoms and molecules are involved and calculate the concentration of solutes in an aqueous solution (the number of solute molecules in a volume of solution).

When carrying out experiments, we use mass to calculate the number of molecules. We must first calculate the **molecular mass**, which is the sum of the masses of all the atoms in a molecule. As an example, let's calculate the molecular mass of table sugar (sucrose), $C_{12}H_{22}O_{11}$, by multiplying the number of atoms by the atomic mass of each element (see the periodic table at the back of the book). In round numbers of daltons, the mass of a carbon atom is 12, the mass of a hydrogen atom is 1, and the mass of an oxygen atom is 16. Thus, sucrose has a molecular mass of $(12\times12)+(22\times1)+(11\times16)=342$ daltons. Because we can't weigh out small numbers of molecules, we usually measure substances in units called moles. Just as a dozen always means 12 objects, a **mole (mol)** represents an exact number of objects: 6.02×1023, which is called Avogadro's number. Because of the way in which Avogadro's number and the unit *dalton* were originally defined, there are 6.02×1023 daltons in 1 g. Once we determine the molecular mass of a molecule such as sucrose, we can use the same number (342), but with the unit *gram*, to represent the mass of 6.02×1023 molecules of sucrose, or 1 mol of sucrose (sometimes called the *molar mass*). To obtain 1 mol of sucrose in the lab, therefore, we weigh out 342 g.

The practical advantage of measuring a quantity of chemicals in moles is that a mole of one substance has exactly the same number of molecules as a mole of any other substance. If the molecular mass of substance A is 342 daltons and that of substance B is 10 daltons, then 342 g of A will have the same number of molecules as 10 g of B. A mole of ethyl alcohol (C_2H_6O) also contains 6.02×1023 molecules, but its mass is only 46 g because the mass of a molecule of ethyl alcohol is less than that of a molecule of sucrose. Measuring in moles makes it convenient for scientists working in the laboratory to combine substances in fixed ratios of molecules.

How would we make a liter (L) of solution consisting of 1 mol of sucrose dissolved in water? We would measure out 342 g of sucrose and then gradually add water, while stirring, until the sugar was completely

dissolved. We would then add enough water to bring the total volume of the solution up to 1 L. At that point, we would have a 1-molar (1M) solution of sucrose. **Molarity**—the number of moles of solute per liter of solution—is the unit of concentration most often used by biologists for aqueous solutions.

Water's capacity as a versatile solvent complements the other properties discussed in this chapter. Since these remarkable properties allow water to support life on Earth so well, scientists who seek life elsewhere in the universe look for water as a sign that a planet might sustain life.

MP3 Tutor

The properties of Water

Hey there, this is Eric Simon and welcome to another MP3 Tutor session. Today's topic is water-a substance that is often taken for granted but has remarkable properties that are essential for life. Water and life are so tied together that biologists find it almost impossible to imagine life without water. Unlike other planets in our solar system, the surface of the Earth is covered with liquid water. In fact, the presence of water is one of the main reasons that Earth is habitable. As you listen to this MP3, NASA has two robot explorers on Mars. Their mission is to search for evidence of liquid water on that planet. Such evidence would be essential to establish the possibility that life might have once existed on Mars. By the way, both rovers did find evidence of liquid water on the surface of ancient Mars, but evidence of life has not yet been found. What is it about water that makes it so essential for life? As with so many other things in biology, the function of water is a consequence of its structure. Taking a minute to understand the structure of water will make it much easier for you to understand why water acts in so many special ways. Probably the first chemical formula you learned was H2O, the formula for water. H2O is a shorthand method of representing that each water molecule is composed of two atoms of hydrogen and one atom of oxygen. One molecule of water has two covalent bonds that join each hydrogen atom to the oxygen atom. Remember that a covalent bond consists of a pair of shared electrons. An oxygen atom has a greater affinity or attraction for electrons than a hydrogen atom does. Therefore, when a hydrogen atom and an oxygen atom share a pair of electrons in a chemical bond, the electrons spend more time around the oxygen nucleus than they do around the hydrogen nucleus. This seems like such a trivial point but it has big consequences. The unequal sharing of electrons creates two separate areas of electrical charge. Around the oxygen atom the molecule is partially negatively charged with more electrons, more of the time. Around the hydrogen atoms the molecule is partially positively charged because the electrons spend less time there. This type of bond is called a polar covalent bond and water is a polar molecule-a molecule with two "poles"-one slightly negative and the other slightly positive. In nature, opposite charges attract. So the positive pole of one water molecule is attracted to the negative pole of another water. In liquid or solid water, the water molecules arrange themselves to take advantage of this attraction between opposite charges. This attraction has a name: hydrogen bonds. Hydrogen bonds tend to

make water sticky, sticking to each other and to other molecules. It is this "stickiness" that underlies the versatile properties of water. Water sticks to all sorts of things, making them wet-cotton jeans, soil, your hair. When water sticks to some other substance, like cotton, it is called adhesion. Water also sticks to itself, as evidenced by a drop of water dripping from a faucet or falling from a cloud. This property also has a name: cohesion. Cohesion is the tendency of water molecules to stick to each other. Now that we understand the basic structure of water, let's talk about four special properties of water. They are: 1 its ability to moderate temperature, 2 the fact that ice floats, 3 its ability to dissolve substances, and 4 surface tension. We'll relate each of these properties back to the sticky structure of water that we just described. The first property of water that we are going to discuss is its ability to moderate temperature. Unlike most common substances, water can absorb a lot of energy with only small changes in temperature. Here's a mind experiment that makes this point: In your imagination, place a pan of water on the stove and turn the burner to high. Wait 30 seconds. Which would more likely burn you at this point: the pan made from steel or the water inside it? You can probably guess that the pan reaches burning temperatures much faster than the water does. This demonstrates that water resists changes in temperature more than steel does. Water's ability to absorb heat is the result of the sticky nature of water. Before the temperature can increase in water some energy is absorbed to overcome the stickiness-to separate water molecules from each other. Energy is therefore put into breaking the hydrogen bonds between waters rather than raising the temperature. A consequence of this property is that water can reduce temperature swings for the planet as a whole and also in specific environments near large bodies of water. Oceans have remarkable abilities to keep the air temperature around them from changing a lot. For example, in Seattle or Boston, which are both along seacoasts, the temperatures rarely get above 100 degrees Fahrenheit or below zero. On the other hand, cities in the U.S. heartland, such as Minneapolis or Omaha, experience these temperatures on a yearly basis. Without the huge oceans, Earth's temperatures would not be compatible with life as we know it. The second property of water that we'll discuss today is the fact that solid water is less dense than liquid water. In other words, ice floats. Do you realize how unique this is? For almost all known substances, the solid form is denser than the liquid form. If you were to make frozen gasoline cubes, for example, the solid gasoline would sink to the bottom of a glass of gasoline. Solid water, or ice, is different because water molecules at freezing temperatures align with each other in a precise arrangement to take advantage of the maximum possible hydrogen bonds. Liquid water, which is not so precisely arranged, is denser than ice. This is almost impossible to describe without a picture so be sure to check out the illustrations in your book and on the website. Why is the fact that ice floats so important to life? Imagine what would happen if ice were more dense than liquid water. During the winter, a layer of ice would form on the top of a lake where it is exposed to the cold air. This ice would sink to the bottom, and more ice would form on top, which would also sink, and so on. Soon enough, the entire lake would be frozen solid from top to bottom and no life could survive the winter. Instead, because ice floats, a thin layer of ice insulates the rest of a lake, allowing aquatic life to survive the winter. Think about that the next time you are clinking ice cubes in your soft drink. The image of

your soft drink brings us to water's third property. Consider what your soft drink is: it's a solution of sugars or sugar substitutes, flavoring, and carbonation. This makes the point that water is a very versatile solvent; that is, water can dissolve many other substances. In fact, water is sometimes called the "universal solvent." Your cells are essentially small packets of water that contain a complex solution of molecules dissolved within the water. Substances that dissolve in water or are attracted by water are called hydrophilic. And guess what? This property is also the result of water "sticking" to hydrophilic substances. Of course, not all substances dissolve in water. Two substances you are familiar with that do not dissolve in water are waxes and oils. Water beads up on waxy surfaces and the oil in salad dressing rises above the watery vinegar. Substances that do not dissolve in water or repel water are called hydrophobic. Both hydrophilic and hydrophobic substances can be found in living systems. The fourth property of water that I'll talk about today is surface tension. You've probably noticed that liquid water seems to have a skin on top. If you are careful, you can suspend a paper clip on top of this skin. This skin is a result of surface tension created when individual water molecules stick to each other. The stickiness creates water droplets or a layer at the surface of a pool or pond of water. This forms a boundary layer that is stout enough to serve as a surface habitat for small insects, snails, and other small organisms. Simple enough, right? Let's recap the topic of water: Water molecules form sticky hydrogen bonds due to their polar covalent bonds. The stickiness of water molecules to each other and to other substances is responsible for properties such as temperature moderation, floating ice, dissolving substances, and surface tension. I hope that this MP3 Tutor has helped you to understand the properties of water. Next time you drink a glass of water show a little appreciation for such a versatile and unique substance. And good luck with the rest of your studying!

Possible Evolution of Life on Other Planets

EVOLUTION Biologists who look for life elsewhere in the universe (known as *astrobiologists*) have concentrated their search on planets that might have water. More than 800 planets have been found outside our solar system, with evidence for the presence of water vapor on a few. In our own solar system, Mars has been a focus of study. Like Earth, Mars has an ice cap at both poles. Images from spacecraft sent to Mars showed that ice is present just under the surface of Mars and that enough water vapor exists in its atmosphere for frost to form. In 2015, scientists found evidence of water flowing on Mars **(Figure 3.10)**, and a study using radar in 2018 concluded there is a large reservoir of liquid water one mile below surface ice. Drilling below the surface may be the next step in the search for signs of life on Mars. If any life-forms or fossils are found, their study will shed light on the process of evolution from an entirely new perspective.

Figure 3.10

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Evidence for liquid water on Mars.
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Dark streaks are visible across the surface of Mars.

Water appears to have helped form these dark streaks that run downhill on Mars during the summer. NASA scientists also found evidence of hydrated salts, indicating water is present. (This digitally treated photograph was taken by the Mars Reconnaissance Orbiter.)